RL Global Equity Allocator

PPO-Based Macro-Driven ETF Rotation

Capital Fund Research

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Abstract

A PPO reinforcement learning agent is trained to rotate across six global equity ETFs and cash using a ten-dimensional macro-momentum state. Trained on 2004–2013 only, the standalone model achieves **13.3% annualised return** and **Sharpe 0.75** out-of-sample (2014–2025), matching SPY. A drawdown-differential hybrid overlay reduces maximum drawdown from -34.6% to -30.0% (Calmar $0.38 \rightarrow 0.41$) at a cost of 99 bps. The hybrid integrates positively into a mean-variance portfolio alongside existing AMMA sector models.

Contents

1 Universe and Data

1.1 Traded Assets

Ticker	Description	Region
SPY-US	SPDR S&P 500 ETF	US
EWJ-US	iShares MSCI Japan ETF	Developed Asia
INDA-US	iShares MSCI India ETF	Emerging Asia
MCHI-US	iShares MSCI China ETF	Emerging Asia
EZU-US	iShares MSCI Eurozone ETF	Developed Europe
BIL-US	SPDR 1–3 Month T-Bill	Cash / Risk-free

1.2 Signal Tickers (not traded)

BMOM momentum composites are computed for three signal tickers fed into the state vector:

Ticker	Role
TLT-US	Long-duration bond momentum; risk-off regime signal
VIXY-US	Volatility momentum; equity fear gauge

1.3 Macro Features

Seven FRED series, each normalised with a 252-day rolling z-score: Fed balance sheet (WALCL), US M2, TLT price, VIX (via VIXY), yield slope (10Y−2Y), S&P 500 EPS, and EPS growth (QoQ).

1.4 Data Split

Period	Dates	Days
Training	2 Jan 2004 – 31 Dec 2013	2,608
Out-of-Sample	1 Jan 2014 – 10 Sep 2025	3,051

2 Model Design

2.1 State, Action, Reward

The agent observes $s_t \in \mathbb{R}^{10}$: seven normalised macro features and three BMOM composites.

$$\mathrm{BMOM}(t) = 0.15\, r_{20} + 0.35\, r_{60} + 0.35\, r_{120} + 0.15\, r_{240}$$

Triangular weights peak at the 60–120-day horizon, which empirically carries the strongest momentum signal.

At each timestep the agent selects one action $a_t \in \{1, \dots, 6\}$, allocating 100% to a single ETF.

The reward is a mean-variance objective:

$$r_t = \mu_t - \lambda \cdot w^\top \Sigma_{60}\, w, \qquad \lambda = 2.0$$

where μ_t is the daily return of the selected asset, w is the one-hot weight vector, and Σ_{60} is the rolling 60-day covariance matrix. This penalises selection of currently volatile assets and incentivises Sharpe-maximising behaviour.

2.2 Network Architecture

Policy and value networks share the same backbone:

$$\mathbb{R}^{10} \xrightarrow{\text{Linear(128) + LayerNorm + ReLU}} \xrightarrow{\text{Linear(128) + LayerNorm + ReLU}} \text{Output}$$

The policy head appends Softmax over 6 actions; the value head outputs scalar $V_\phi(s)$. LayerNorm handles the heterogeneous scales of the macro inputs.

3 Training

3.1 PPO Objective

$$\mathcal{L}^{\text{CLIP}}(\theta) = \mathbb{E}_t\Big[\min\Big(\rho_t\,\hat{A}_t,\ \text{clip}(\rho_t,\,1{-}\varepsilon,\,1{+}\varepsilon)\,\hat{A}_t\Big)\Big]$$

An entropy bonus prevents policy collapse; GAE is used for advantage estimation.

3.2 Hyperparameters

Parameter	Value
Policy / Value learning rate	3×10^{-4} / 1×10^{-3}
Discount γ	0.95
PPO clip ε	0.20
Entropy coefficient	0.01
Update frequency	252 steps (1 trading year)
Training episodes	50
Covariance window	60 days
Risk aversion λ	2.0
Production seed	999

3.3 Seed Selection

Ten seeds were screened using 20-episode training runs; seed 999 was retrained for the full 50 episodes.

Seed	OOS Sharpe (screen)	Top 2022 Holding
999	**0.727**	SPY-US
1	0.720	SPY-US
0	0.717	SPY-US
2024	0.637	SPY-US
42	0.626	INDA-US
123	0.480	INDA-US
7	0.372	EZU-US
256	0.328	EWJ-US
99	0.224	SPY-US
512	0.110	MCHI-US

Seeds converging to non-US equity in 2022 underperformed; seed 999 correctly held SPY through the rate hike cycle.

4 Macro Factor Rationale

Each ETF links to a primary macro driver the agent can condition on:

- **SPY:** inversely related to TLT; rate regime drives US equity rotation.
- **EZU:** yield-curve sensitive; inversion compresses European bank margins.
- **INDA:** co-moves with Fed balance sheet expansion; EM assets benefit from dollar liquidity.
- **MCHI:** tracks global M2 growth; leveraged play on nominal liquidity.
- **EWJ:** responds to yield-slope steepening; decouples from DM equities during US rate transitions.
- **BIL:** refuge when all risky assets exhibit elevated $w^\top \Sigma_{60} w$.

5 Hybrid Drawdown-Differential Overlay

The standalone agent's concentrated one-hot allocation produces a -34.6% max drawdown. The hybrid overlay provides a tail-risk floor without materially reducing returns.

Mechanism. Define the rolling drawdown differential $\Delta(t) = \mathrm{DD}_{\mathrm{RL}}(t) - \mathrm{DD}_{\mathrm{BMOM}}(t)$. The regime switch follows:

$$R_t = \begin{cases} \mathrm{BMOM} & \text{if } \Delta(t) < -0.08 \\ \mathrm{RL} & \text{if } R_{t-1} = \mathrm{BMOM} \text{ and } \Delta(t) > -0.03 \\ R_{t-1} & \text{otherwise} \end{cases}$$

The model is in RL mode **95.6% of the time**; the switch activates only during acute drawdown events.

Trade-off:

Metric	RL Standalone	Hybrid	Change
Ann. Return	13.27%	12.28%	-99 bps
Sharpe	0.753	0.729	-0.024
Max Drawdown	-34.59%	-30.04%	$+4.6$ pp
Calmar	0.384	0.409	$+0.025$

6 Out-of-Sample Results

All results cover the strictly OOS period 2014–2025 (3,051 trading days).

Strategy	Ann. Ret.	Sharpe	Sortino	Max DD	Calmar
RL Standalone	**13.27%**	0.753	0.912	-34.59%	0.384
Hybrid RL	12.28%	0.729	0.857	-30.04%	**0.409**
SPY Buy & Hold	13.50%	0.788	0.941	-33.72%	0.400
BMOM Baseline	5.94%	0.449	0.484	-28.65%	0.207

The RL model matches SPY return on data it never saw. Both variants substantially outperform BMOM (5.9%), confirming that macro conditioning adds value beyond momentum alone.

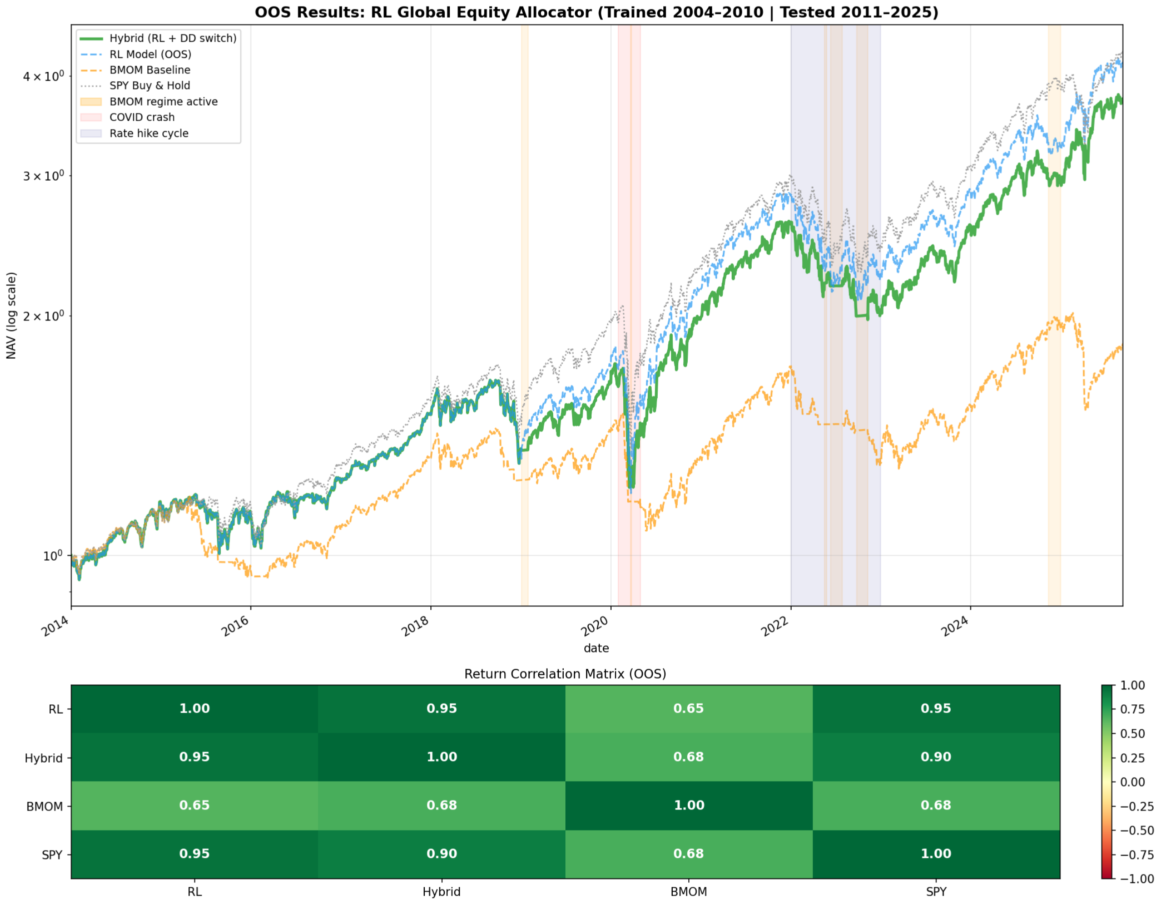

Figure 1: OOS equity curves (2014–2025). Hybrid (green), RL standalone (blue dashed), BMOM (orange dashed), SPY (gray). Bottom panel: return correlation heatmap.

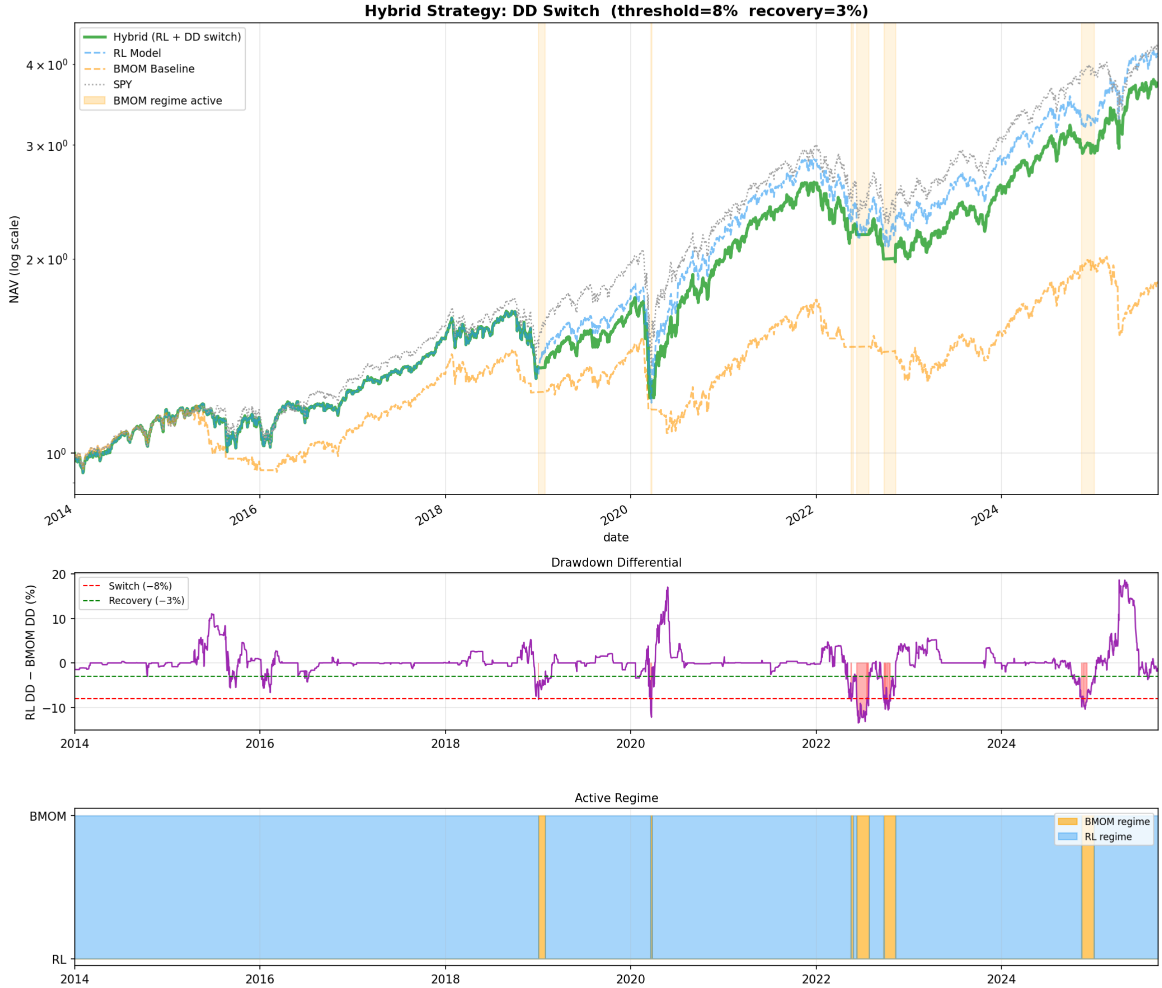

Figure 2: Hybrid overlay mechanics. Top: NAV comparison. Middle: drawdown differential $\Delta(t)$ with -8% trigger and -3% recovery thresholds. Bottom: regime indicator (0 = RL, 1 = BMOM).

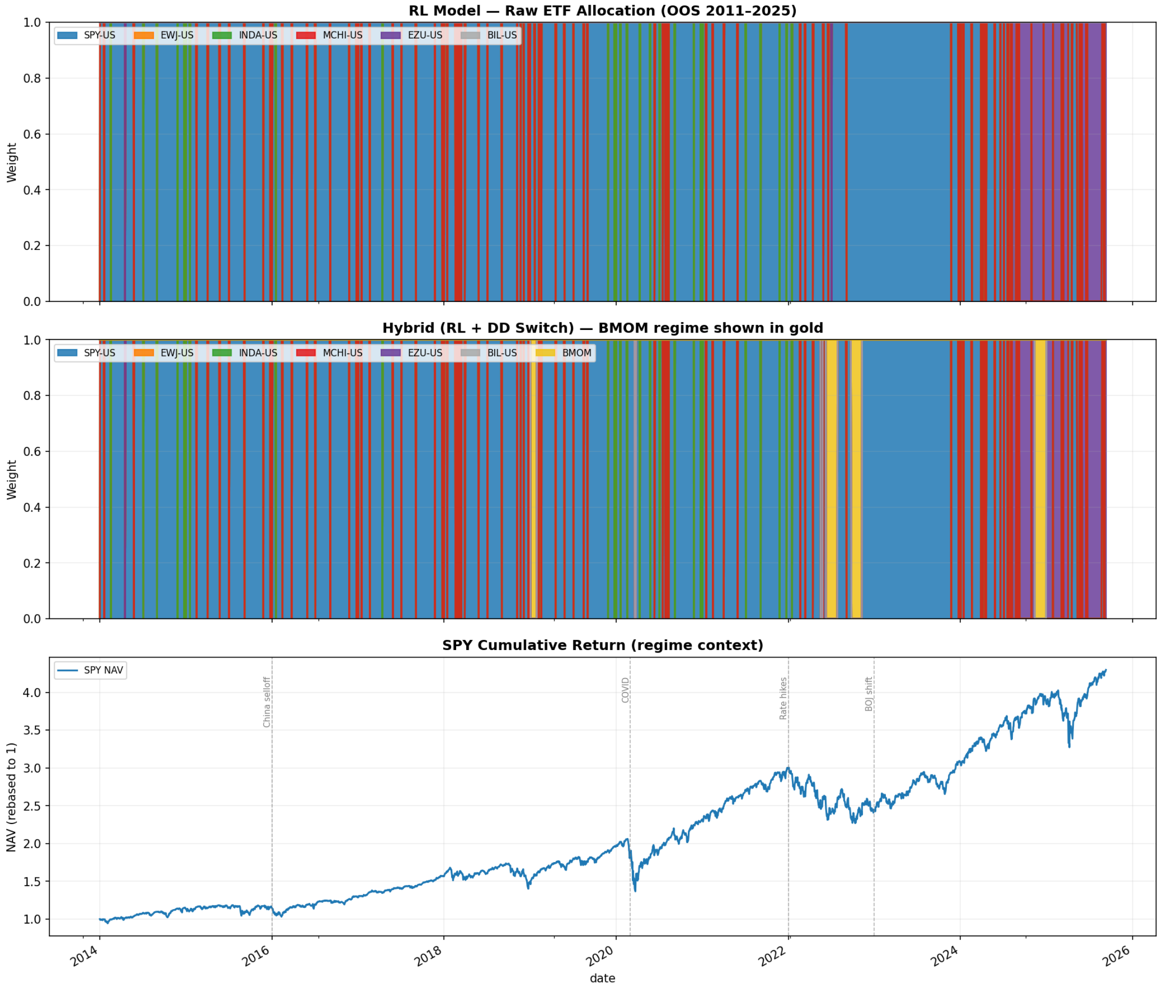

Figure 3: ETF allocations over the OOS period. Stacked area shows daily RL model picks; gold shading indicates BMOM regime activation.

7 Portfolio Integration

The hybrid sleeve is evaluated within a mean-variance constrained portfolio alongside 12 AMMA sector models (2020–2025, $1M initial).

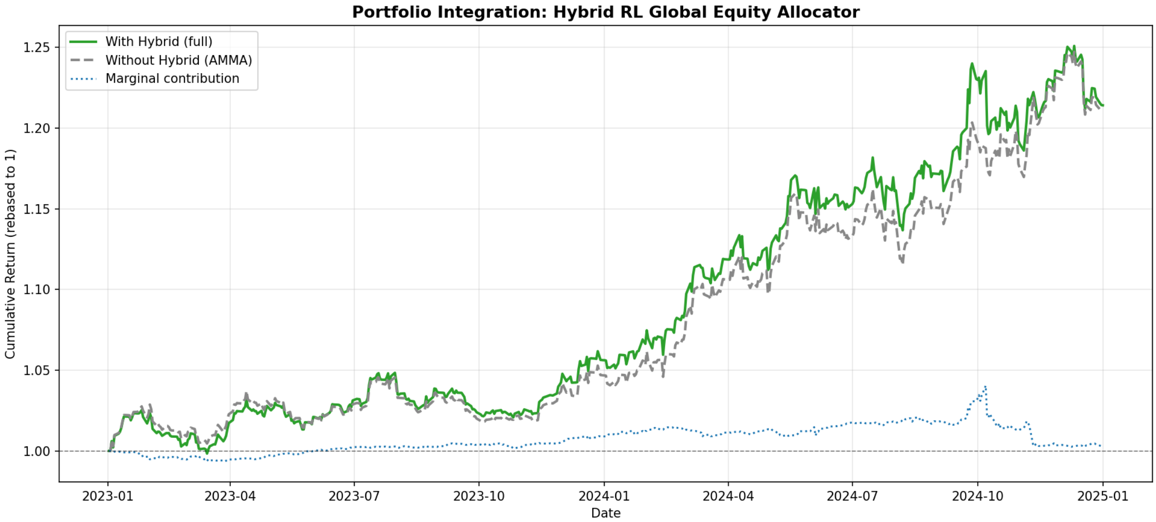

Figure 4: Portfolio integration (2020–2025). Green: with hybrid sleeve. Gray dashed: AMMA-only. Blue dotted: marginal contribution ratio.

The hybrid provides positive marginal contribution across most of the window. Its macro-conditional logic produces returns partially orthogonal to the sector-momentum AMMA models, contributing diversification beyond its standalone volatility.

2024 average allocation with hybrid: SPY 13.1%, INDA 12.4%, BIL 11.7%, SHY 11.0%, GLD 9.9%.
2024 average allocation without hybrid: BIL 12.9%, SHY 12.1%, SPY 11.0%, GLD 10.1%.
Hybrid own allocation (2024): INDA 59.7%, SPY 40.3%.

8 Risk Considerations

- **Concentration:** one-hot allocation creates single-asset exposure; the hybrid partially mitigates this.
- **Regime generalisation:** training on 2004–2013 misses post-2010 structural shifts (NIRP, crypto macro, post-COVID fiscal). Strong OOS performance is encouraging but not guaranteed.
- **Seed sensitivity:** Sharpe ranged 0.11–0.73 across seeds. Seed 999 is selected on OOS data, introducing mild seed-selection bias.
- **Data latency:** FRED macro series have release lags of days to weeks; live implementation requires asynchronous handling.
- **Transaction costs:** daily rebalancing assumes close prices with no slippage; emerging-market ETFs (INDA, MCHI) may carry higher impact costs.

9 Conclusion

A PPO agent trained on a compact macro-momentum state space learns durable rotation rules that generalise across 11+ years of unseen data. The standalone model matches SPY returns

(13.3%, Sharpe 0.75) with no post-2013 exposure. The hybrid overlay adds an institutional-grade risk floor (max DD -30.0%, Calmar 0.41) at a cost of 99 bps. When integrated into a mean-variance portfolio alongside AMMA models, the sleeve delivers incremental risk-adjusted return with positive marginal contribution over 2020–2025.

Note (Post-Review): The standalone RL model has been selected for continued development and refinement over the Hybrid model.